RONALD KATENDE, PhD

Applied Mathematics | Numerical Analysis | Scientific Computing | Scientific Machine Learning

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Academic Profile

Applied mathematician working in numerical analysis, scientific computing, partial differential equations, mathematical modelling and scientific machine learning. My research asks: *What information and computation are sufficient for a reliable scientific decision?*

I study this through stability, inverse problems, numerical linear algebra, sparse sensing, reduced-order methods and reliable learning under limited information and computation.

Research Programme and Interests

Computational mathematics and numerical analysis. Partial differential equations, finite-element and finite-difference methods, numerical linear algebra, stability, conditioning, approximation and iterative methods.

Scientific machine learning and inverse problems. Physics-informed neural networks, operator and inverse learning, sparse sensing, low-rank and reduced-order methods, uncertainty quantification and condition-aware reconstruction.

Learning, computation and representation. Information requirements for reliable learning and recovery; task-relative numerical linear algebra; computational work and span; representation complexity; and mathematical limits of learning architectures.

Dynamical systems and scientific modelling. Differential-equation models, nonlinear dynamics, epidemic systems, scientific forecasting and computational modelling, particularly under limited data, sensing and computational resources.

Academic Appointments

2026–present — *Lecturer, Department of Mathematics, Kabale University, Uganda*
Undergraduate and postgraduate teaching; independent research; postgraduate supervision; undergraduate research supervision; curriculum development; and departmental and faculty service.

2019–2026 — *Assistant Lecturer, Department of Mathematics, Kabale University, Uganda*
Teaching in mathematics, statistics and scientific computing; student research supervision; research; and postgraduate teaching from 2025.

2016–2019 — *Assistant Lecturer, Uganda Christian University, Uganda*
Undergraduate teaching in mathematics, statistics, scientific computing and quantitative methods.

Education

2025 — *PhD in Mathematics, Makerere University, Uganda*
Specialisation in Scientific Computing and Machine Learning. Research collaboration at the University of Bergen, Norway. Thesis: *Numerical Analysis of Physics-Informed Neural Networks for Solving PDEs.*

2017 — *MSc in Mathematics, Pan African University / JKUAT, Kenya*
Dissertation: *On the Location of a Free Boundary for American Options.*

2014 — *BSc in Mathematics and Statistics, Makerere University, Uganda*
First Class Honours.

Peer-Reviewed and Accepted Publications

1. Katende, R. Non-Asymptotic Stability and Consistency Guarantees for Physics-Informed Neural Networks via Coercive Operator Analysis. *Communications in Nonlinear Science and Numerical Simulation* 159, 109909 (2026). doi:10.1016/j.cnsns.2026.109909.
2. Katende, R. (2026). Information contraction and stability under inverse-stable memory clocks in data-driven learning. *Research in Mathematics*, 13(1), 2743449. doi.org/10.1080/27684830.2026.2743449
3. Katende, R. Rethinking Data-Efficient Artificial Intelligence for Low-Resource Settings. *Machine Learning with Applications* 23, 100796 (2026). doi:10.1016/j.mlwa.2025.100796.
4. Katende, R., Kasumba, H., Kakuba, G., & Mango, J. M. On the Error Bounds for ReLU Neural Networks. *IAENG International Journal of Applied Mathematics* 54(12), 2602–2611 (2024). Article link.
5. Katende, R., Kasumba, H., Kakuba, G., & Mango, J. M. A Proof of Convergence and Equivalence for 1D Finite Element Methods and ReLU Neural Networks. *Annals of Mathematics and Computer Science* 25, 97–111 (2024). doi:10.56947/amcs.v25.392.
6. Mango, J. M., Katende, R., & Kasumba, H. Adaptive Error-Bounded Hierarchical Matrix Compression for Efficient and Accurate Physics-Informed Neural Networks. *Engineering Letters* 33(5), 1605–1622 (2025). Article link.
7. Namugera, F., Katende, R., Kurama, O., & Magero, J. M. Long-Range Epidemic Spreading with Aging in Structured Networks. *Physica A: Statistical Mechanics and its Applications* 676, 130841 (2025). doi:10.1016/j.physa.2025.130841.
8. Katende, R. A Data-Efficient Framework for Analysing Structural Transformation in Low- and Middle-Income Economies. *Scientific Reports* 15, 34216 (2025). doi:10.1038/s41598-025-15952-3.
9. Katende, R. Bridging the Macro–Micro Divide through a New Paradigm for Climate Resilience Assessment in Data-Scarce Regions. *PLOS Climate* 5(1), e0000662 (2026). doi:10.1371/journal.pclm.0000662.
10. Katende, R. Modeling Resource Conflict and Inequality in Fragile Systems: A Spatial Agent-Based Framework with a Uganda Case Study. *PLOS ONE* 21(9), e0357283 (2026). doi:10.1371/journal.pone.0357283.
11. Katende, R. Cross-Country Comparative Analysis of Climate Resilience and Localized Mapping in Data-Sparse Regions. *Frontiers in Environmental Science* 13, 1495950 (2025). doi:10.3389/fenvs.2025.1495950.

Selected Conference Activity

DeepMath 2026	Accepted poster: *When Can Depth Replace Width? A Sharp ReLU Depth–Width Frontier with Persistent State.*
SIAM Applied Linear Algebra 2027	*Task-Relative Numerical Linear Algebra: Same Spectrum, Different Work*; contributed lecture submission.
FLICS 2026	Two papers accepted for presentation, Valencia, Spain.

Research Funding and Funded Projects

- *Kabale University Research and Innovation Fund*, Awardee, 2025–2026.
- *Government of Uganda Research and Innovation Fund*, Co-Investigator, 2022–2024.
- *Government of Uganda Research and Innovation Fund*, Co-Investigator, 2023–2025.

Awards and Fellowships

- *TWAS Research Award*, 2025.
- *Computational Geometry Fellowship*, University of Agder, Norway, 2024.
- *Mathematics for Africa Research Prize*, European Mathematical Society, 2023.
- *CIMPA Fellowship*, 2023.

Postgraduate Research Supervision

- Two MSc students successfully completed under my supervision/co-supervision; three additional MSc researchers supervised or co-supervised in applied mathematics, scientific computing and mathematical modelling.
- Current postgraduate projects include differential-equation modelling, physics-informed neural networks, inverse parameter estimation, dynamical systems and computational mathematics.
- Invited to contribute to PhD supervision in mathematical and computational research projects.
- Supervised or guided more than 50 undergraduate research projects in mathematics, statistics, modelling and computational methods.

Teaching Experience

Numerical & Applied Mathematics	Numerical Analysis; Advanced Numerical Methods (PG); Differential Equations; Partial Differential Equations; Numerical Linear Algebra; Linear Programming.
Analysis & Linear Algebra	Linear Algebra; Real Analysis; Functional Analysis.
Probability, Statistics & Data	Probability and Statistics; Statistical Modelling; Data Analytics; Applied Data Structures.
Programming & Scientific Computing	Python Programming (UG/PG); R Programming; scientific-computing instruction for mathematics and statistics students.

- Regular teaching load of more than *11 credit units per semester.*
- Teach across undergraduate and postgraduate programmes in mathematics, statistics, numerical methods and scientific computing.
- Delivered intensive Python training in Statistical Modelling for MSc Statistics students.
- Delivered a one-month Python and scientific-computing course for MSc Computational Mathematics students at Makerere University.
- Experience in student assessment, research-project supervision, course development and computational laboratory instruction.

International Research Experience and Collaboration

University of Bergen, Norway	Research collaboration in scientific machine learning and numerical analysis during doctoral research.
Area Science Park, Trieste, Italy	Three-month research visit in 2025 focused on sparse optimisation and computation under data scarcity.
Justus Liebig University Giessen, Germany	Invited research visit and academic seminar, 2025.
University of Havana, Cuba	Invited academic visit and seminar, 2023.
University of Milan	Research collaboration and academic engagement.
University of Rwanda	Institutional and academic engagement involving Kabale University and the University of Rwanda.

Selected Research Prototypes and Applied Work

AMMED	Multi-timescale memory methods for early detection of clinical deterioration under explicit false-alarm constraints, with transfer testing across independent hospital populations.
GridTwin Sentinel	Low-compute early-warning and decision-support prototype for electricity infrastructure combining forecasting, change detection, temporal memory, physical consistency checks, network localisation and uncertainty.
Industrial digital twins	Sparse-sensor full-field reconstruction, condition-aware sensor selection and cross-regime validation for high-dimensional physical systems.
Open research practice	Reproducible numerical experiments, version-controlled scientific code and research-software development across mathematical, machine-learning and scientific-computing projects.

Selected Research Contributions

Scientific machine learning. Stability and consistency analysis for physics-informed neural networks, with links to finite-element methods.

Task-relative computation. Mathematical criteria for when global matrix descriptors fail to capture the work required by a specific numerical task.

Sparse recovery and sensing. Condition-aware reconstruction from limited measurements, including sensor selection and reduced-order methods.

Learning under limited information. Information, memory, computation and measurement limits for reliable learning and recovery.

Applied mathematical modelling. Applications to epidemics, climate, conflict, infrastructure, digital twins and clinical early warning.

Academic Leadership and Programme Development

- Mathematics representative on the Faculty of Science Research Committee, Kabale University.
- Coordinator of Innovations and Exhibitions, Department of Mathematics, Kabale University.
- Developed and submitted a proposed academic programme in Computational Science and Industrial Analytics.
- Represented Kabale University at the National Council for Higher Education Innovation Exhibition.
- Contribute to curriculum development, postgraduate supervision, departmental research development and academic service.

Service to the Research Community

***Journal peer review**: Scientific Data*; *Neurocomputing*; *Machine Learning: Science and Technology*; *International Journal of Machine Learning and Cybernetics*; *International Journal of Data Science and Analytics*; *Letters in Spatial and Resource Sciences*; *Discover Artificial Intelligence*; *Discover Sustainability*; *Future Business Journal*; *Journal of Machine Learning and Data Science*.

***Conference reviewing**:* ITU Kaleidoscope; Africa at NeurIPS 2026 Workshop (three research submissions reviewed).

Technical Competencies

Scientific computing	Python, MATLAB, R, NumPy, SciPy, PyTorch, scikit-learn, Git, Linux and LaTeX.
Numerical computation	Numerical simulation, optimisation, numerical linear algebra, differential-equation solvers, inverse computation and scientific visualisation.
Research practice	Reproducible computational experiments, research-software development, version-controlled workflows and preparation of technical and mathematical research outputs.

References

Academic referees available upon request and supplied directly where required by the recruiting institution.